

Problem 1 - Simulation of Sources from Monopoles

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Figure 1 shows the spherical spreading from a monopole. The frequency is 1 kHz. The sound pressure level has been calculated at two distances: 1 m and 4 m.

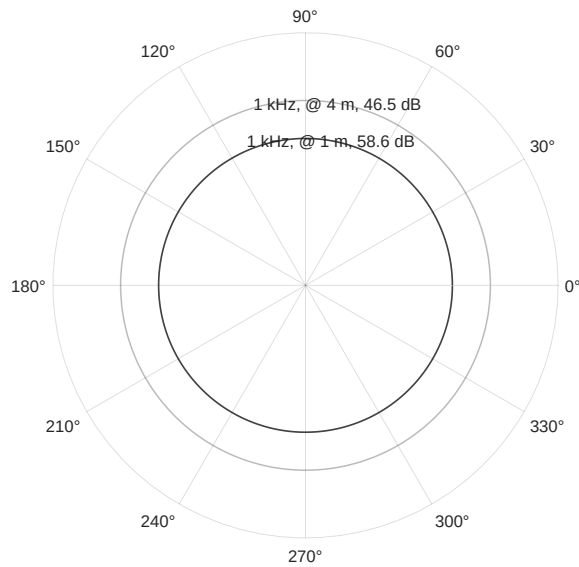


Figure 1: Monopole spherical spreading for 1 kHz. The radius of the polar plot is the sound pressure level.

Figure 2 shows the sound pressure level as a function of distance for a 1 kHz monopole source.

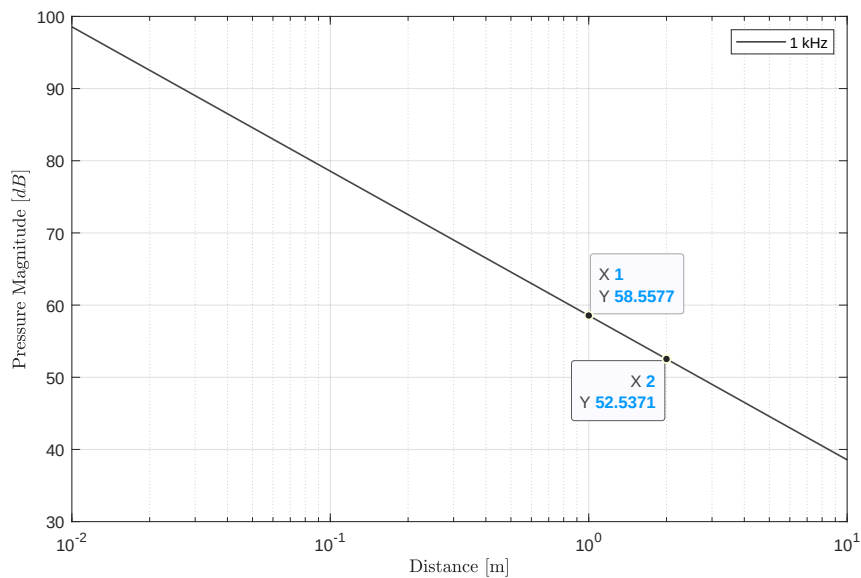


Figure 2: Monopole pressure drop versus distance for 1 kHz.

The pressure drop is proportional to $\frac{1}{r}$; doubling the distance yields -6 dB of level change (see data points).

Problem 1b

Figure 3 illustrates the directivity patterns for a monopole, dipole, lateral quadrupole, and a linear quadrupole for 1 kHz. The receivers were placed at a 1 m distance around each of the sources.

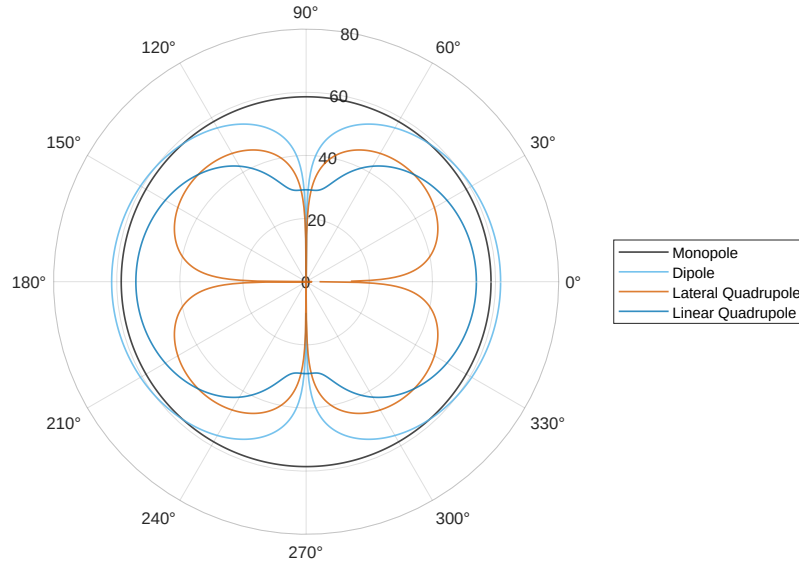


Figure 3: Directivity patterns for a 1 kHz monopole, dipole, lateral quadrupole, and linear quadrupole.

Table 1 lists the sound pressure levels for the 4 types of patterns for 1 kHz.

Source Type	Sound Pressure Level [dB]
Monopole	58.6
Dipole	61.6
Lateral Quadrupole	48.2
Linear Quadrupole	53.9

Table 1: Sound pressure levels for 1 kHz at 1 m distance.

The monopole had a volume velocity of $1.4 \frac{\text{m}^3}{\text{s}}$.

The dipole had two equal strength, in-phase elements with a separation of 0.0429 m, which is less than the 1 kHz wavelength of 0.343 m. The volume velocity for both sources was $1.4 \frac{\text{m}^3}{\text{s}}$.

The lateral quadrupole had four equal strength, diagonally opposite in-phase pairs with a separation of 0.0429 m. The volume velocity for both sources was $1.4 \frac{\text{m}^3}{\text{s}}$.

The linear quadrupole had three sources, the end sources were equal strength and in-phase, while the center source was twice the strength of the single sources and 180° out-of-phase. The separation between all three sources was 0.0429 m. The volume velocity of the end sources was $1.4 \frac{\text{m}^3}{\text{s}}$ and the volume velocity for the center source was $2.8 \frac{\text{m}^3}{\text{s}}$.

Problem 1c

Figure 4 shows a set of directivity patterns for a 9-element line source with a separated of 0.0429 m centered at the origin along the x-axis (demarcated by the 180° to 0° line in the polar plot) for 100 Hz, 2 kHz, and 5 kHz. The receivers are placed at a 1 m distance around the line source.

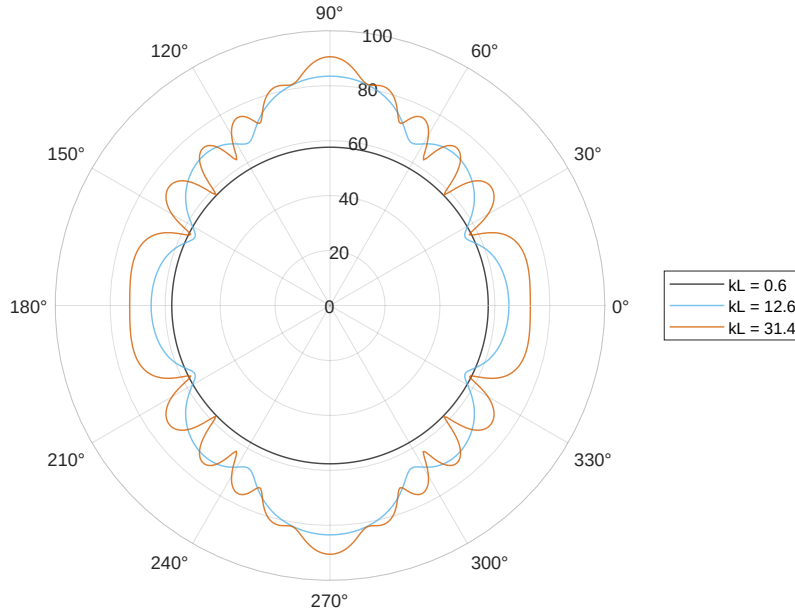


Figure 4: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with small source separation.

The length of the array is 0.343 m, or the 1 kHz wavelength. As the frequency of the source increases (i.e., $k \cdot L$), the patterns have larger directivity but with more side lobes.

Figure 5 shows the directivity patterns for a similar line source, but with the sources now separated by 0.172 m.

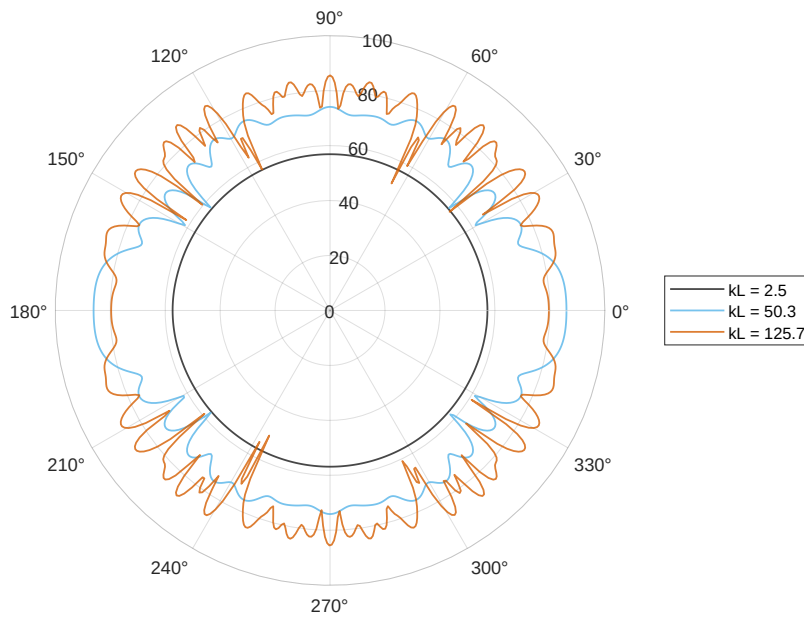


Figure 5: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with large source separation.

The length of the array is now 1.37 m, or 4 times the 1 kHz wavelength. As the frequency of the source increases (i.e., $k \cdot L$), the patterns have larger directivity but with more side lobes. Here, unlike the previous case, there are a greater number of side lobes for the 2 kHz signal (blue curve).

Problem 1d

Figure 6 shows the monopole source locations in the xy-plane. The number of sources in the plane was 100 and “a” was set to 1.715 m (5 times the 1 kHz wavelength of 0.343 m).

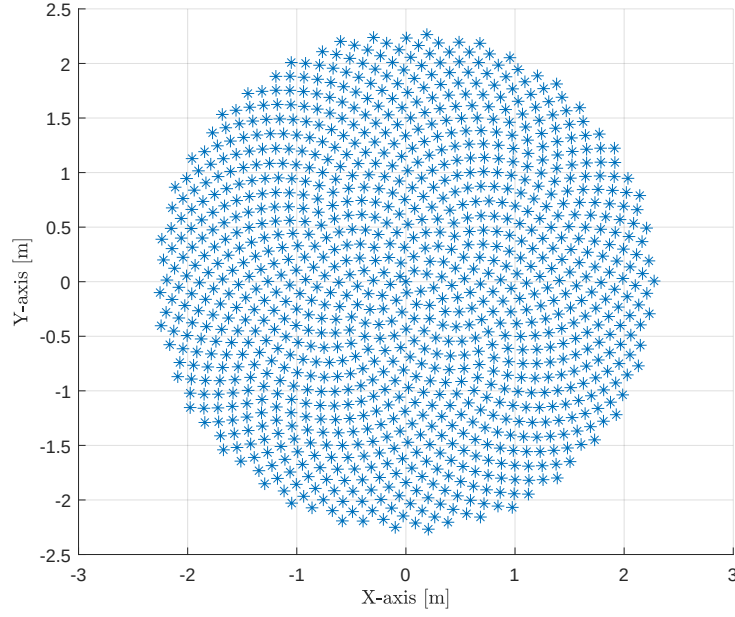


Figure 6: Monopole source locations in the xy-plane.

Figure 7 shows the estimated and theoretical null locations along the z-axis above the xy-plane for the baffled piston.

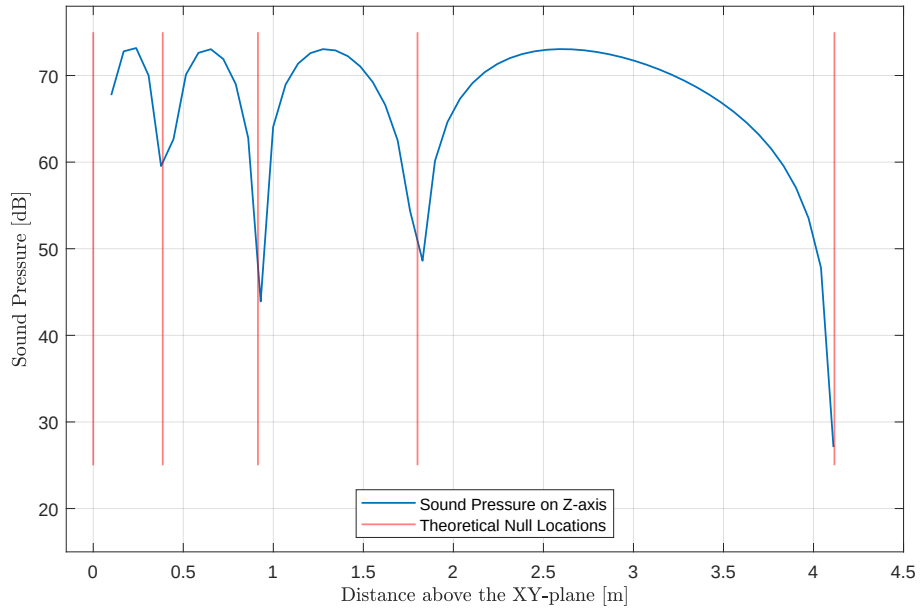


Figure 7: The estimated and theoretical null locations along the z-axis for the baffled piston.

The factor $\frac{a}{\lambda}$ was 5; there are 5 theoretical null locations. There is good agreement between the estimated and theoretical null points. The estimated null near the surface of the piston is governed by starting receiver location on the z-axis. As this distance is made smaller, the sound pressure level grows rapidly due to the calculation method of the monopole sources.

Problem 2 - Active Control of Enclosure Opening

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

Figure 1 shows the sound power level reduction versus octave band frequency for the linear quadrupole system.

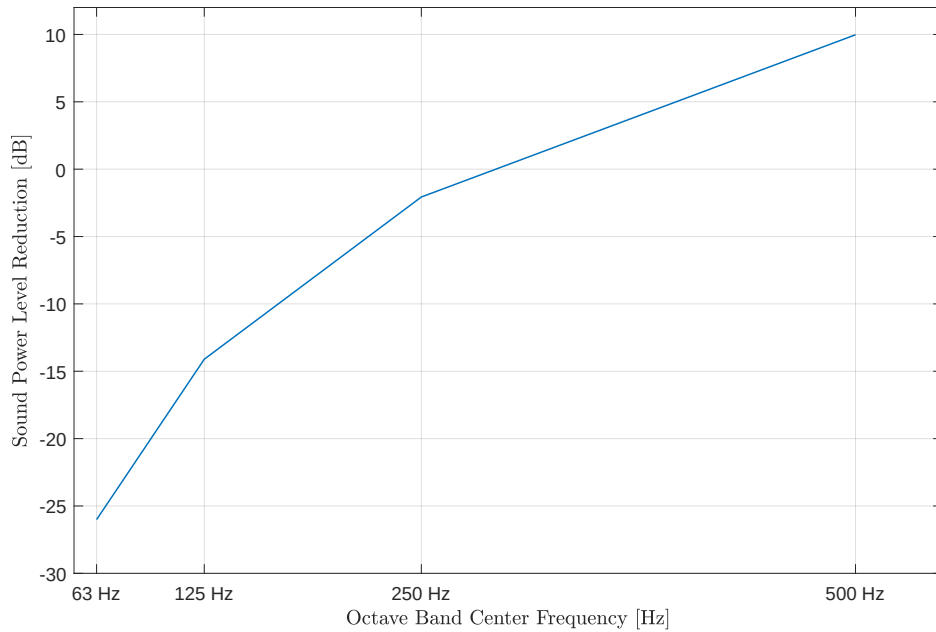


Figure 1: Sound power level reduction as a function of octave band center frequency.

add intuitive description of what is happening

Problem 2b

Figure 2 shows the sound power level reduction versus octave band frequency for the linear quadrupole system.

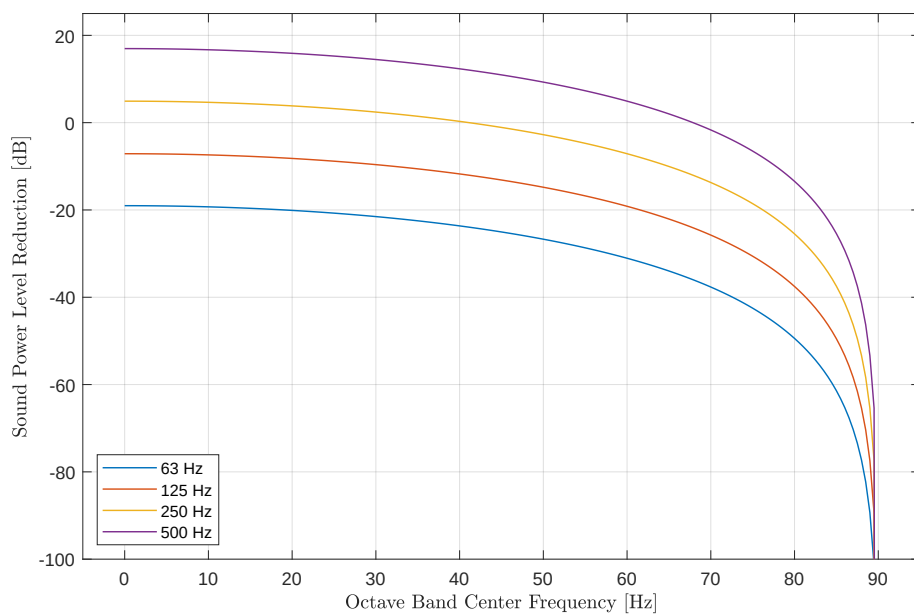


Figure 2: Sound power level reduction as a function of octave band center frequency.

add intuitive description of what is happening

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

Part b - The three speakers have inde
The Matlab code for this problem is listed in Appendix 3.

Part 3a

Figure ?? illustrates the dynamic vibration absorber (DVA) system.
Table 1 lists the stiffness and damping coefficients for the system.

M1	15	kg
K1	0	$\frac{N}{m}$
C1	0	$\frac{kg}{s}$

M2	110	kg
K2	9,810	$\frac{N}{m}$
C2	1,962	$\frac{kg}{s}$ ¹

M3	0	kg
K3	1e5	$\frac{N}{m}$
C3	1e4	$\frac{kg}{s}$ ²

Table 1: Stiffness and damping coefficients for the DVA system.

Damping was implemented by adding with an imaginary component to the respective spring stiffness³.

¹0.2 fraction of respective spring stiffness.
²0.1 fraction of respective spring stiffness.
³Not equivalent to using a damping vector in the argument to the given Matlab nDOF function script-file.

Part 3b

Figure ?? shows the admittance response of the original system and DVA system based on the equations presented in the tutorial lecture on DVA system analysis.

The washing machine admittance at 300 RPM is reduced from the blue curve resonance peak to a point between the two resonances on the red curve.

Part 3c - Admittance

Figure ?? shows the admittance response of the DVA system using the nDOF function script-file. The stiffness of the coupling spring, k_3 , is set to place the null of the admittance response at the 300 RPM operating state of the washing machine. *The admittance at 300 RPM is approximately $1.2 \times 10^{-6} \frac{\text{m}}{\text{N}}$* ⁴

Part 3d - Displacement

Figure ?? shows the displacement of the washing machine as a function of RPM. The ramp up to 300 RPM occurs quickly so that the large displacements are transient and the maximum speed is 350 RPM (the large displacement associated with the resonance peak at higher frequencies is not reached).

The displacement at 300 RPM is approximately $38.2 \times 10^{-6} \text{ m}$.

Part 3e

Figure ?? shows the force applied to the floor by the washing machine.

The force applied to the floor at 300 RPM is approximately $3.75 \times 10^{-3} \text{ N}$.

⁴This is theoretical and not realistic in practical terms, but the admittance is reduced.

Problem 4 - Washing Machine Abuse Testing

The Matlab code for this problem is listed in Appendix [4](#).

Part 4a

1 Appendix - Matlab Code for Problem 1

2 Appendix - Matlab Code for Problem 2

3 Appendix - Matlab Code for Problem 3

4 Appendix - Matlab Code for Problem 4